



REPORT WEEK 4

Coffee Shop Sales Analysis



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Coffee Shop Sales — Profitability, Forecast & Strategy Report

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Dataset size (cleaned): **149,116 rows, 80 products, 9 categories**

1) Objectives

1. Clean and standardize the sales dataset.
2. Quantify **revenue & profit** at product and category levels.
3. Forecast short-term sales to anticipate demand.
4. Provide **actionable recommendations** to increase profits and reduce losses.

2) Methodology (high-level)

- **Cleaning:** normalized headers; coerced dates/prices/qty; trimmed text; removed exact duplicates; filtered invalid rows (non-positive qty, missing date).
- **Derived fields:**
 - $\text{total_sales} = \text{transaction_qty} \times \text{unit_price}$
 - Profit model (assumption-based; see below): $\text{profit} = \text{total_sales} - (\text{COGS} + \text{overhead})$
- **Aggregations:** product & category rollups; daily & monthly trends.
- **Forecast:** 30-day daily revenue forecast (Holt-Winters if available, otherwise a robust moving-average fallback).
- **Outputs:** cleaned data, summary tables, insights, recommendations.

3) Profit Model — Assumptions (transparent & tunable)

Because the raw file had **no unit cost**, I used conservative coffee-shop norms:

- **COGS% by category**
 - Drinks (Coffee/Tea/Chocolate): **35%**

- Bakery/Food: **40%**
- Beans/Packaged/Retail: **55%**
- Merchandise: **65%**
- Fallback (unknown): **45%**
- **Overhead allocation: 0.75** per transaction (labor, utilities, etc.).

These assumptions are clearly separable — if you have your actual COGS and overhead, plug them in and the rankings update deterministically.

4) Data Quality & Cleaning Snapshot

- Rows after cleaning: **149,116**
- Unique products: **80** | Categories: **9**
- Date range: **2023-01-01 → 2023-06-30**
- No structural anomalies after filtering; unit prices and quantities are in valid ranges.

5) Key Results

5.1 Business totals (assumption-based profit)

- **Total revenue: Rs 698,812.33**
- **Total profit: Rs 327,950.52**
- Overall profit margin (modeled): **~46.9%**

5.2 Monthly revenue trend

- Best month: **2023-06 — Rs 166,485.88**
- Worst month: **2023-02 — Rs 76,145.19**
- Clear upward trend from March → June; suggests seasonality/awareness effects.

5.3 Category performance (top 5 by revenue)

- **Coffee — Revenue Rs 269,952.45, Profit Rs 131,657.09, Margin 48.8%**
- **Tea — Revenue Rs 196,405.95, Profit Rs 93,577.12, Margin 47.6%**
- **Bakery — Revenue Rs 82,315.64, Profit Rs 32,292.38, Margin 39.2%**

- **Drinking Chocolate** — Revenue **Rs 72,416.00**, Profit **Rs 31,227.80**, Margin **43.1%**
- **Coffee beans** — Revenue **Rs 40,085.25**, Profit **Rs 24,740.66**, Margin **61.7%** ← standout margin driver

5.4 Top 10 products by profit (assumption-based)

1. **Latte Rg** — Rev **Rs 19,112.25**, Profit **Rs 10,250.96**, Margin **53.6%**
 2. **Sustainably Grown Organic Lg** — **Rs 21,151.75**, **Rs 9,412.71**, **44.5%**
 3. **Cappuccino Lg** — **Rs 17,641.75**, **Rs 9,388.14**, **53.2%**
 4. **Dark chocolate Lg** — **Rs 21,006.00**, **Rs 9,281.55**, **44.2%**
 5. **Morning Sunrise Chai Lg** — **Rs 17,384.00**, **Rs 9,177.10**, **52.8%**
- 6–10. Similar strong earners include popular large hot drinks (detailed in workbook).

Interpretation: Large-size hot beverages dominate profit — high ticket + healthy margins.

5.5 Bottom 10 products by profit (assumption-based)

- **Caramel syrup** — Rev **Rs 2,060.80**, Profit **Rs –160.31**, Margin **–7.8%**
- **Chocolate syrup** — **Rs 2,126.40**, **Rs –130.23**, **–6.1%**
- **Hazelnut syrup** — **Rs 1,897.60**, **Rs –97.07**, **–5.1%**

Remaining low contributors are mostly **add-ons/condiments** with small revenue that get penalized by the per-transaction overhead.

6) Forecast (30-day daily revenue)

- Short-term outlook: **Stable to mildly rising** vs. late-June baseline.
- Operational implication: **Maintain inventory** for top sellers (large hot drinks, beans) and **staff** appropriately for peaks implied by late-week patterns.